Data Analysis with R and Rust

Descriptive Analytics and Data Analysis

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2026-III





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Outline

1. Descriptive Analytics
2. Data Analysis





Outline

1 Descriptive Analytics

2 Data Analysis





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R for Descriptive Statistics

what happened?

R Language

R is a **statistical programming language** designed for data analysis, **visualization**, and **statistical computing**, widely used in *academia*, *biostatistics*, and *data science*.

Core capabilities:

- Vectors and data frames as native first-class types.
- Built-in functions: `mean()`, `sd()`, `summary()`, `cor()`, and `lm()`.

Key packages:





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~ pandas





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matplotlib





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Key packages:

- `dplyr`: data manipulation and filtering grammar.
- `tidyr`: reshaping and pivoting data frames.
- `psych`: descriptive stats, correlation, and factor analysis.
- `caret` / `tidymodels`: ML model training and evaluation pipelines.

pandas + spacy





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scikit-learn

numpy + scipy





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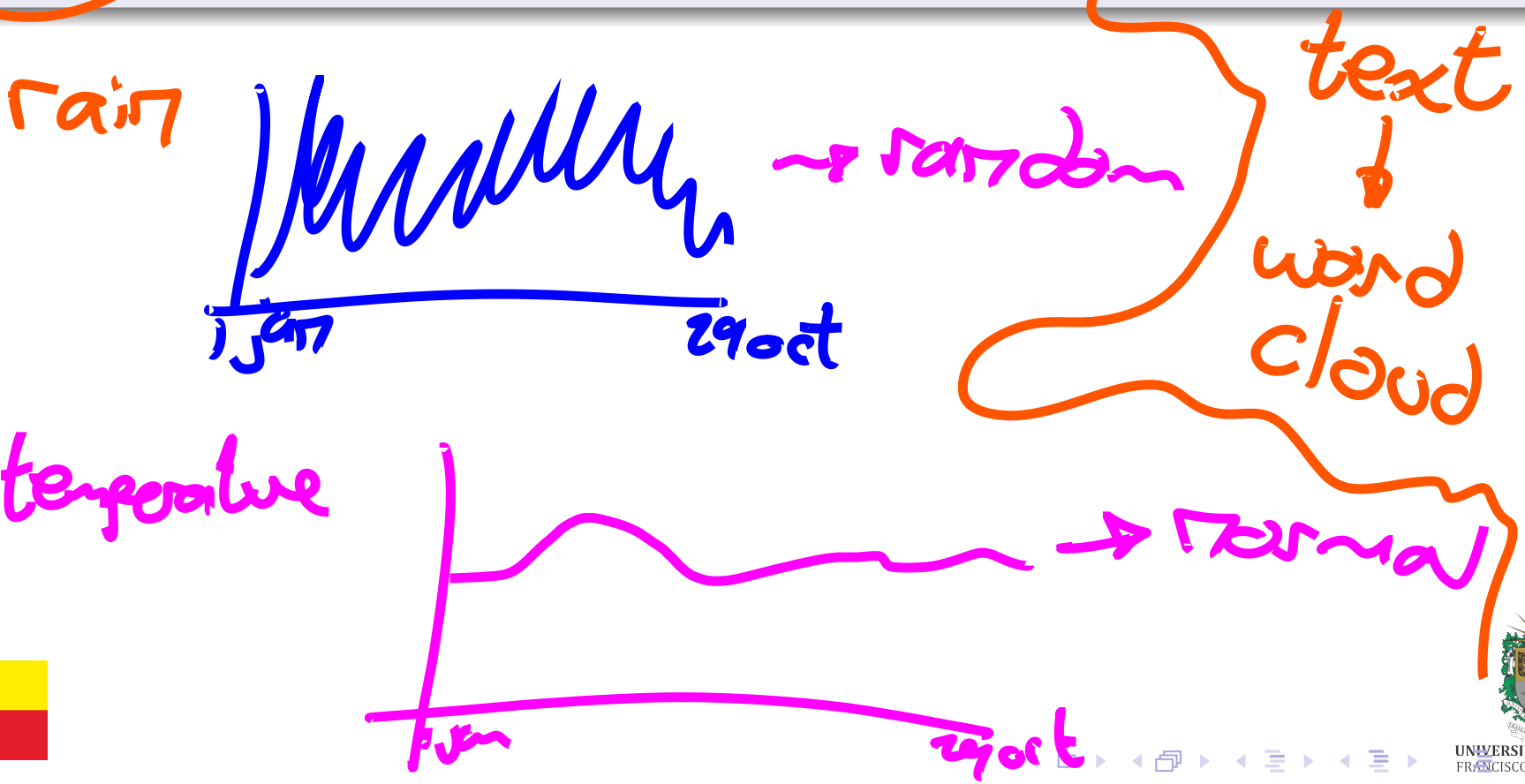




Data Composition I

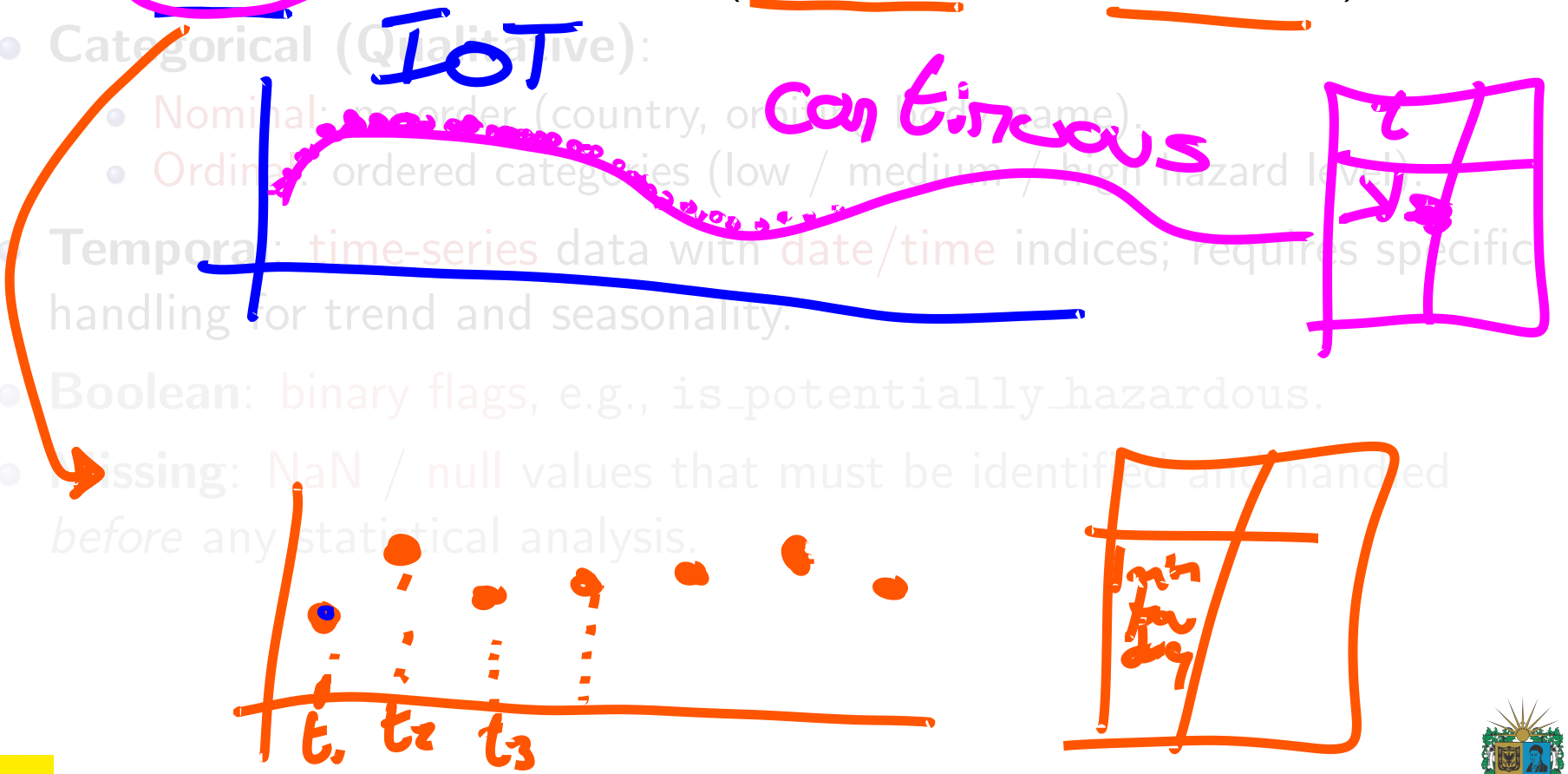
Data Types in a Dataset

Understanding data composition means identifying **what variable types** exist and how they are distributed, which determines which *statistical methods* are applicable.



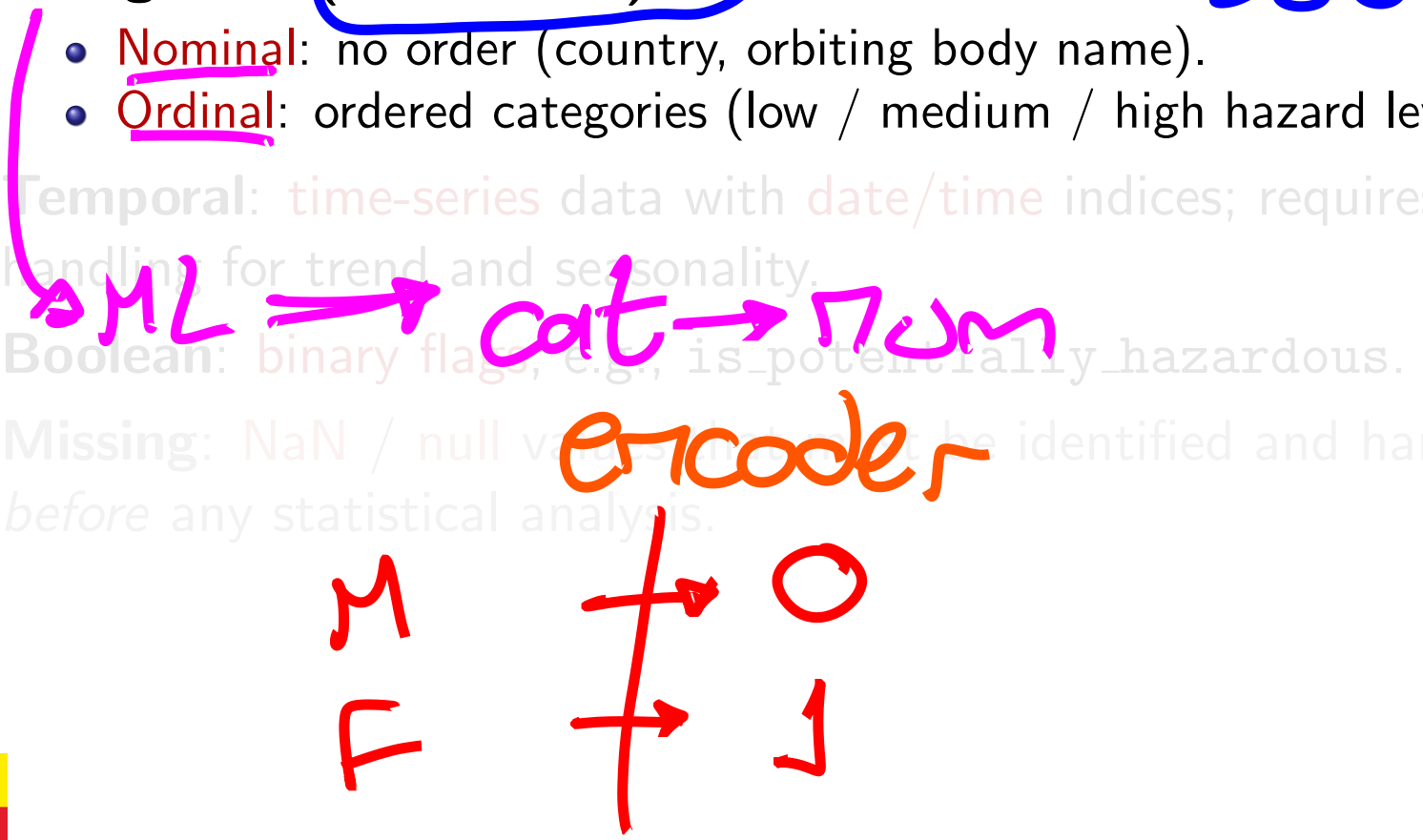
Data Composition II

- **Numerical (Quantitative)**:
  - Continuous: any real value in a range (velocity, distance, temperature).
  - Discrete: countable integers (asteroid count, events per day).
- **Categorical (Qualitative)**:
  - Nominal: no order (country, orbit type, name).
  - Ordinal: ordered categories (low / medium / high hazard level).
- **Temporal**: time-series data with date/time indices; requires specific handling for trend and seasonality.
- **Boolean**: binary flags, e.g., `is_potentially_hazardous`.
- **Missing**: NaN / null values that must be identified and handled *before* any statistical analysis.



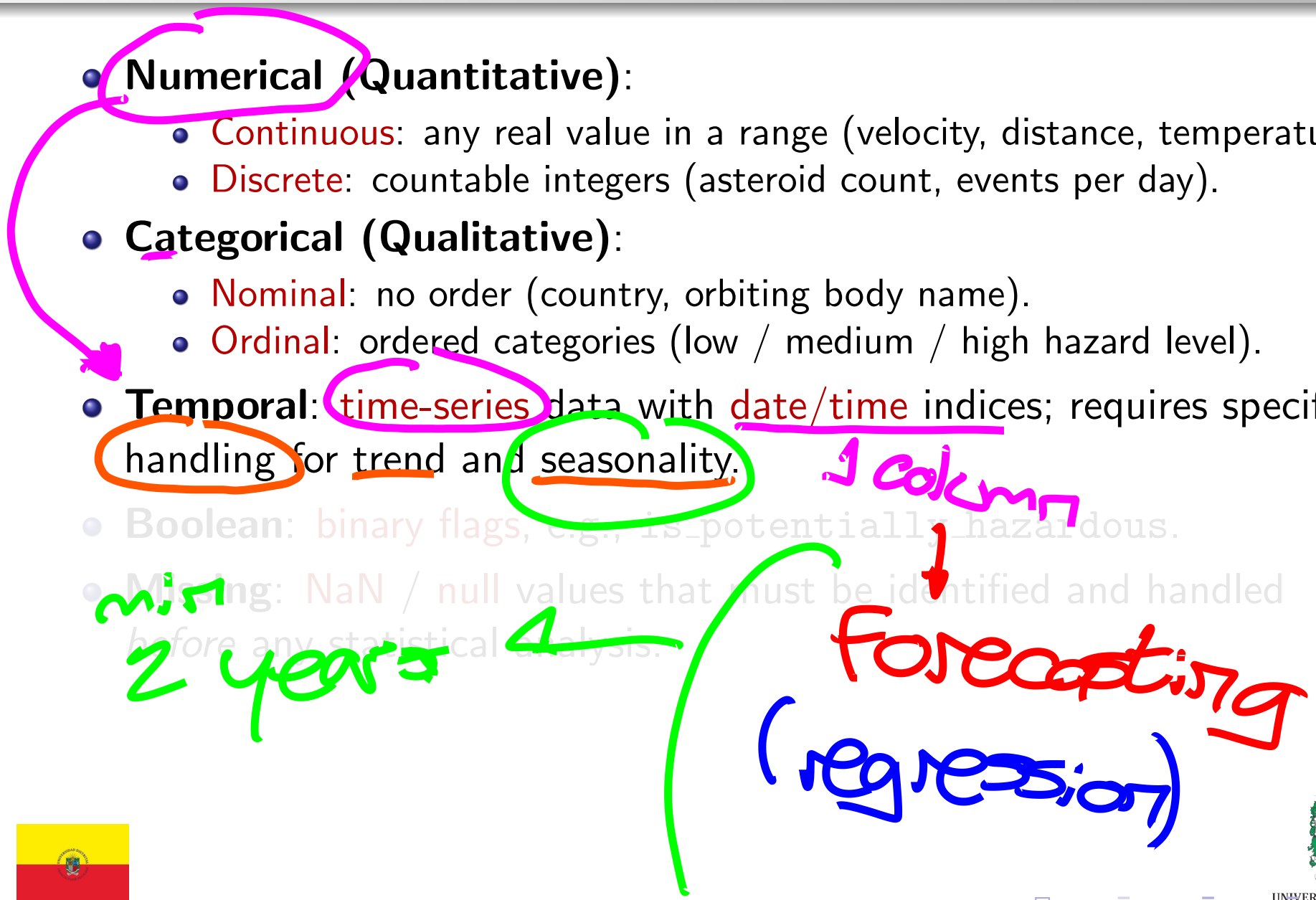
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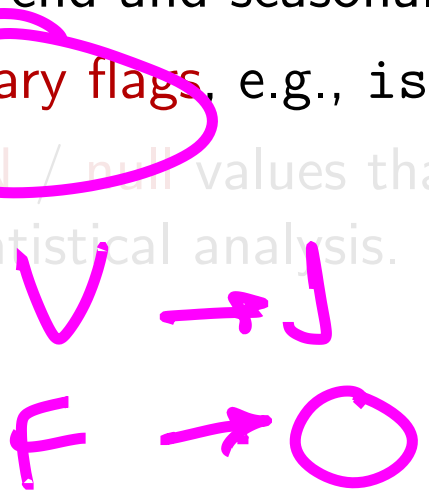




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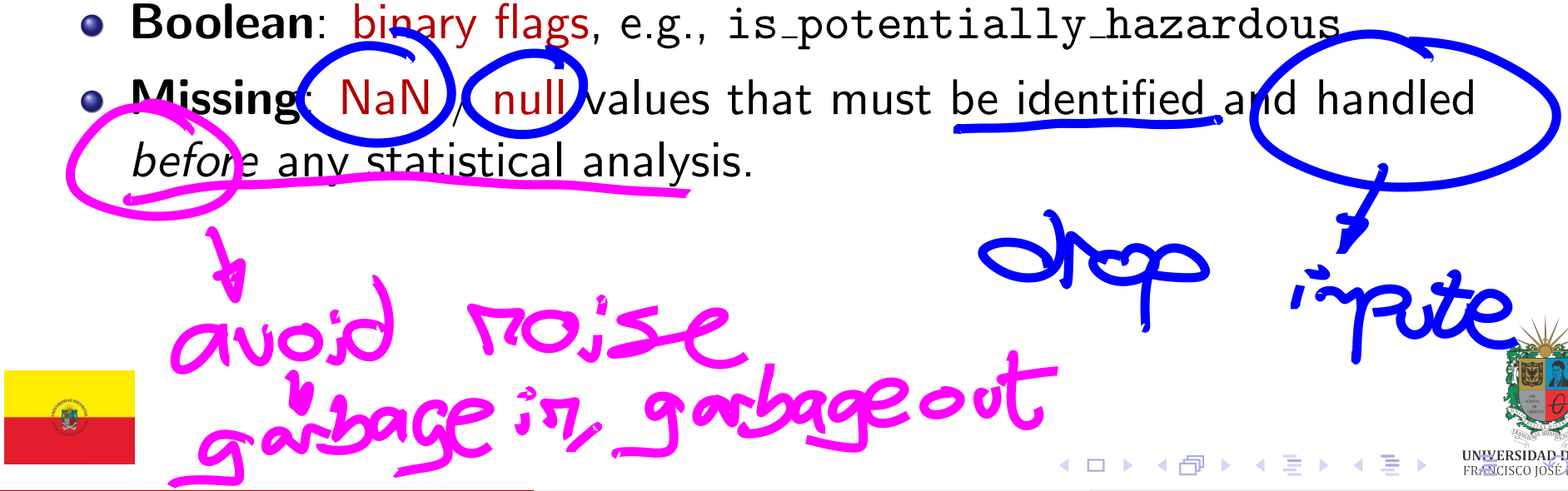






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Mean and Median

Mean (Arithmetic Average)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Sensitive to **outliers**; best for symmetric distributions.

Median

The middle value when data is sorted in ascending order.

Robust to outliers; preferred for **skewed** distributions.





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mean = 4

median = 3

mean = 10.8

median = 1





Mode and Range

Mode

The most frequent value in the dataset.

Useful for categorical and **discrete** data; a dataset can be unimodal, bimodal, or multimodal.

Range

$$R = x_{\max} - x_{\min}$$

Simplest measure of spread; **highly sensitive** to extreme values.





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Normal Data, Standard Deviation, and Variance

Normal (*Gaussian*) Distribution

A symmetric, **bell-shaped** continuous probability distribution fully described by its mean μ and standard deviation σ:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

Variance

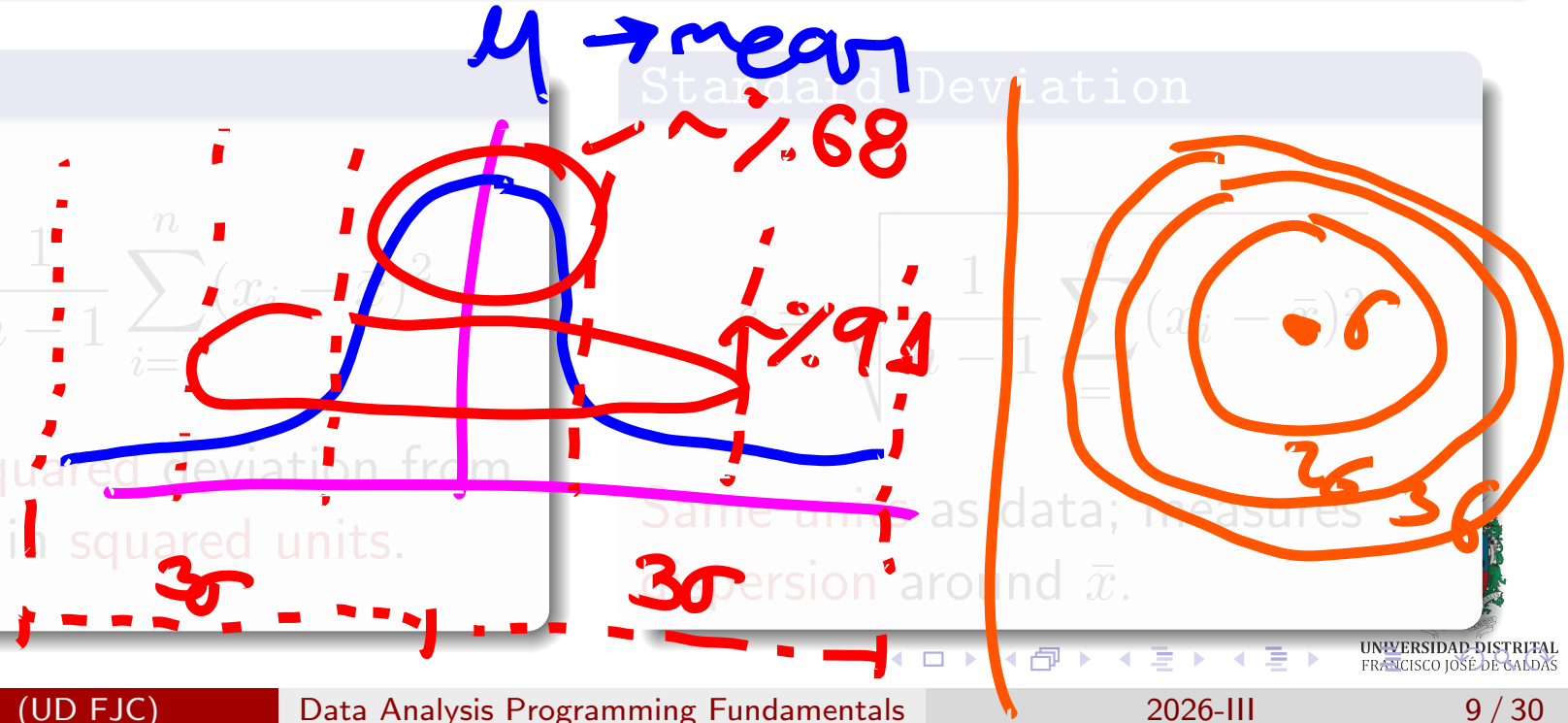
$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Average squared deviation from the mean; in squared units.

Standard Deviation

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Same units as data; measures dispersion around $\bar{x}$.



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Skewness and Kurtosis

Skewness

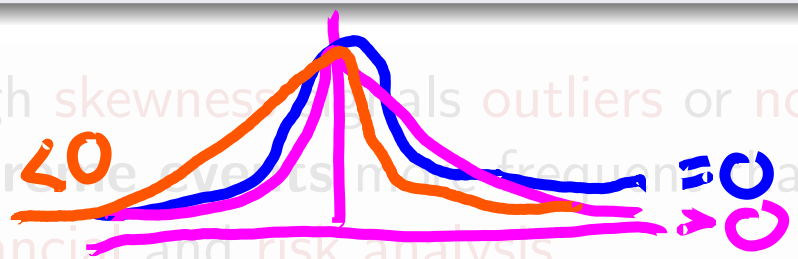
$$\gamma_1 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3$$

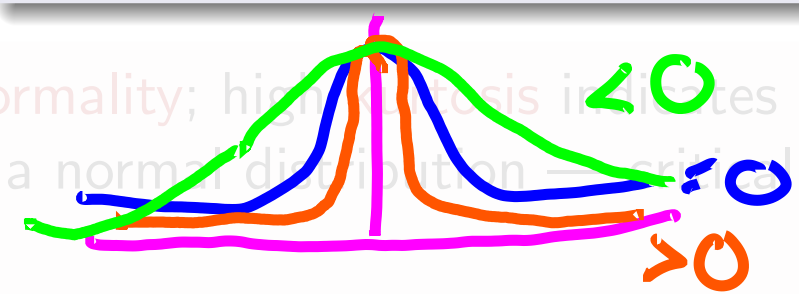
- $\gamma_1 = 0$: symmetric (normal).
- $\gamma_1 > 0$: right-skewed, long right tail.
- $\gamma_1 < 0$: left-skewed, long left tail.

Excess Kurtosis

$$\gamma_2 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^4 - 3$$

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High skewness signals outliers or non-normality; high kurtosis indicates extreme events more frequent than in a normal distribution — critical for financial and risk analysis.

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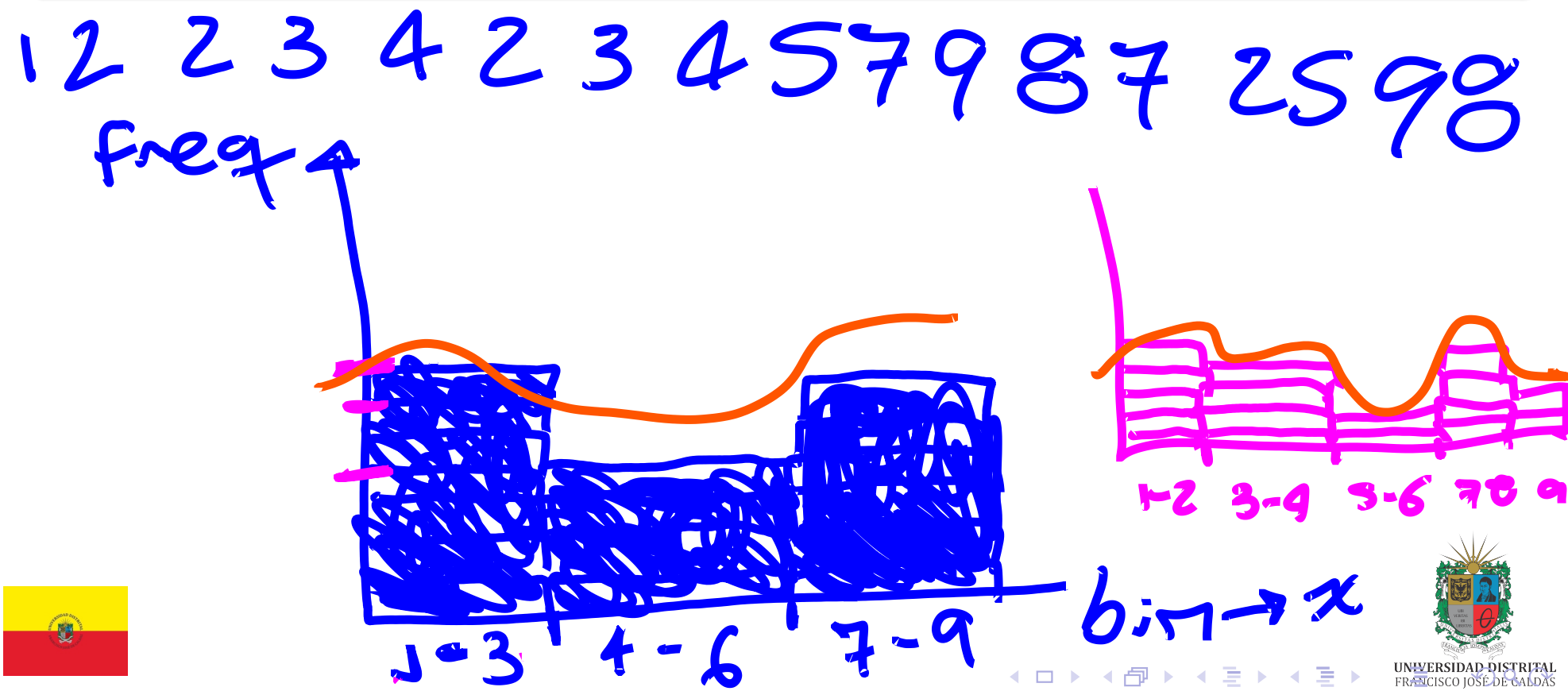
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Histograms and Data Distributions

Histogram

A histogram divides data into contiguous **bins** and plots the frequency or density per bin, revealing the *shape*, *spread*, and *modality* of the data distribution.



Histograms and Data Distribution [II]

- **Bin width**: too narrow ⇒ noisy; too wide ⇒ over-smoothed.
  Sturges' rule: $k = \lceil 1 + \log_2 n \rceil$ bins for n observations.
- **Unimodal**: one peak — possibly normal. **Bimodal**: two peaks — suggests a mixture of populations.
- **Density plot (KDE)**: smooth continuous estimate of distribution using a kernel function; area under curve $= 1$.
- **Frequency vs. Density**:
  - Frequency: raw count of observations per bin.
  - Density: count $\div$ (total $\times$ bin width) — sums to 1.
- **Interpretation**: shape reveals skewness, outliers, and whether normality is a reasonable assumption.





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Box Plots and Data Confidence I

Box Plot (Box-and-Whisker)

A boxplot summarizes a distribution with **five statistics**: $\min$, Q_1, Q_2 (median), Q_3, $\max$, and visually exposes outliers and distributional spread.

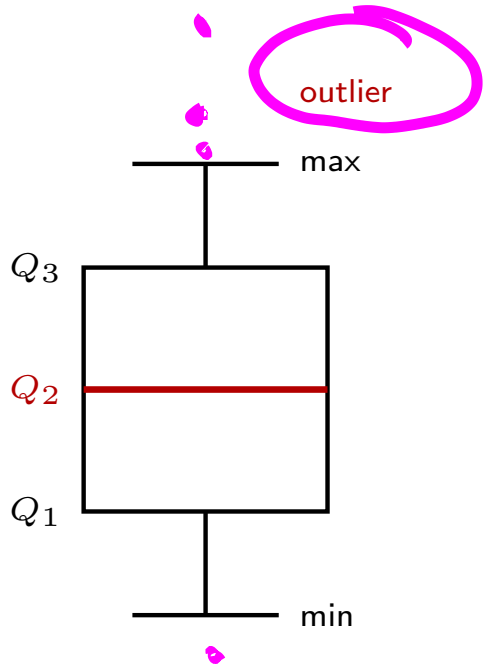
- $Q_1 = 25$th percentile; $Q_3 = 75$th percentile.
- $IQR = Q_3 - Q_1$: the interquartile range (box width).
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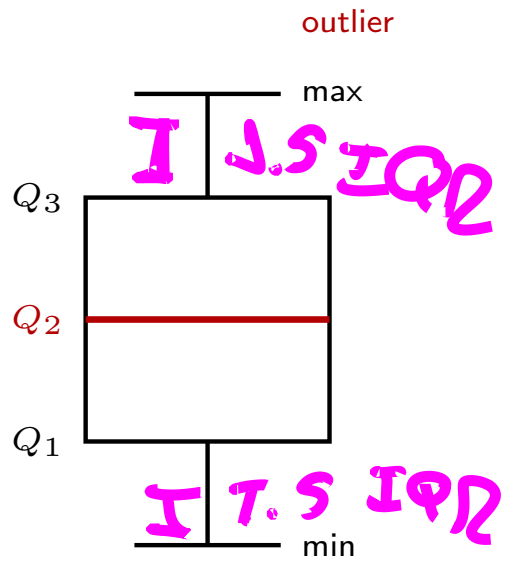


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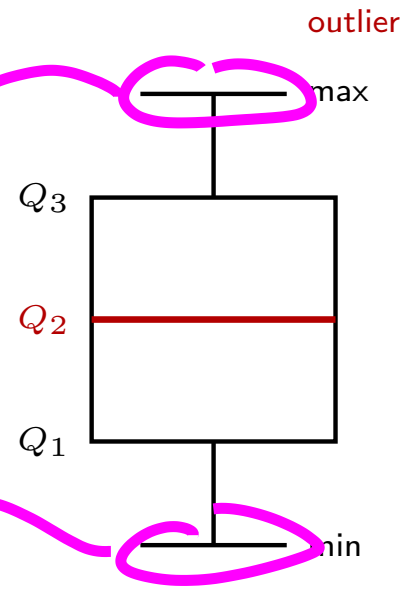


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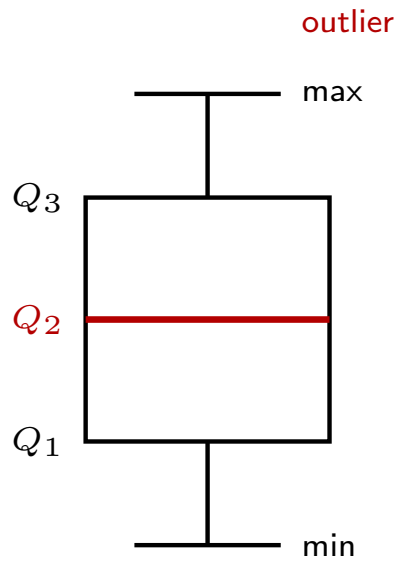


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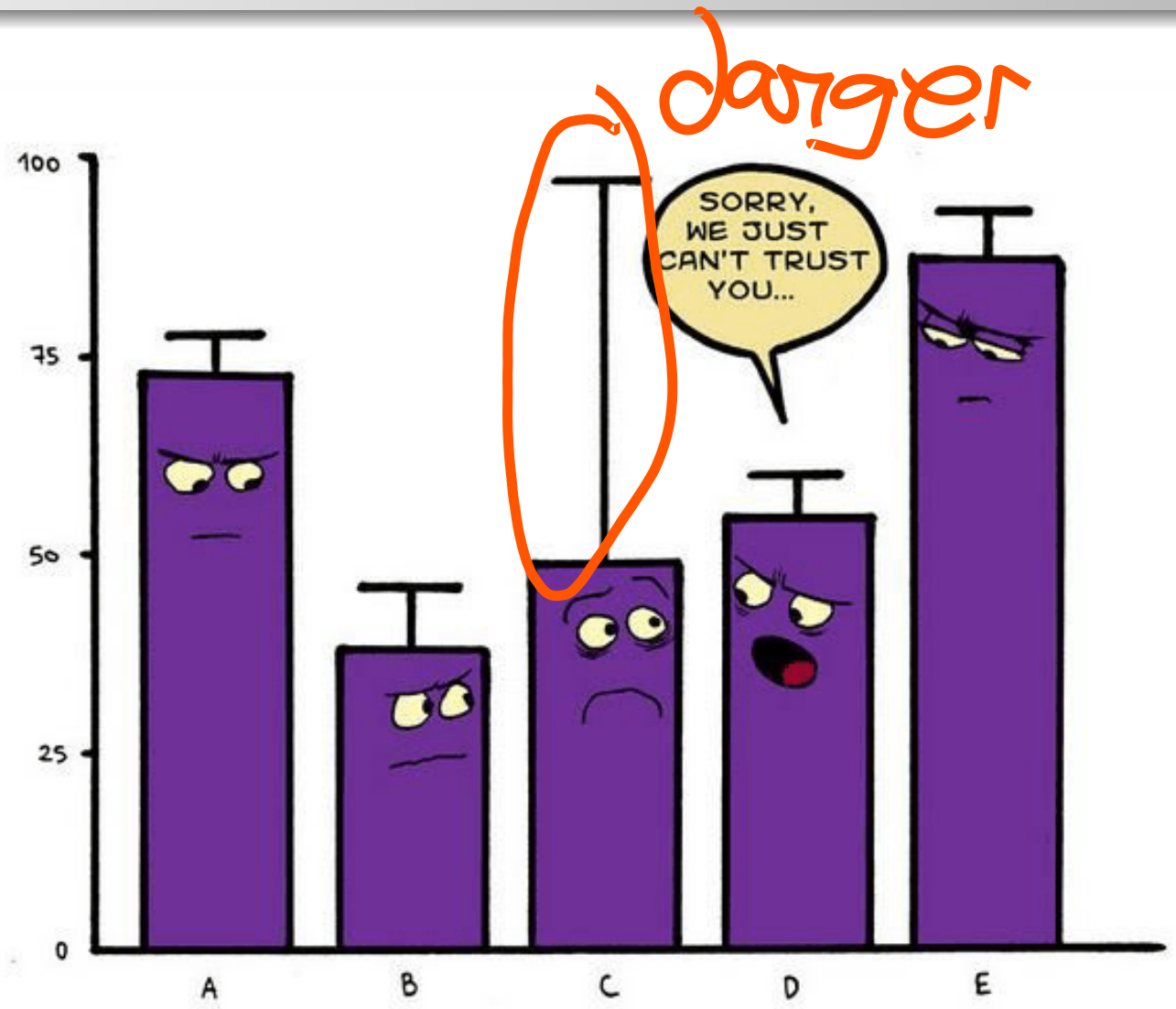
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p-value

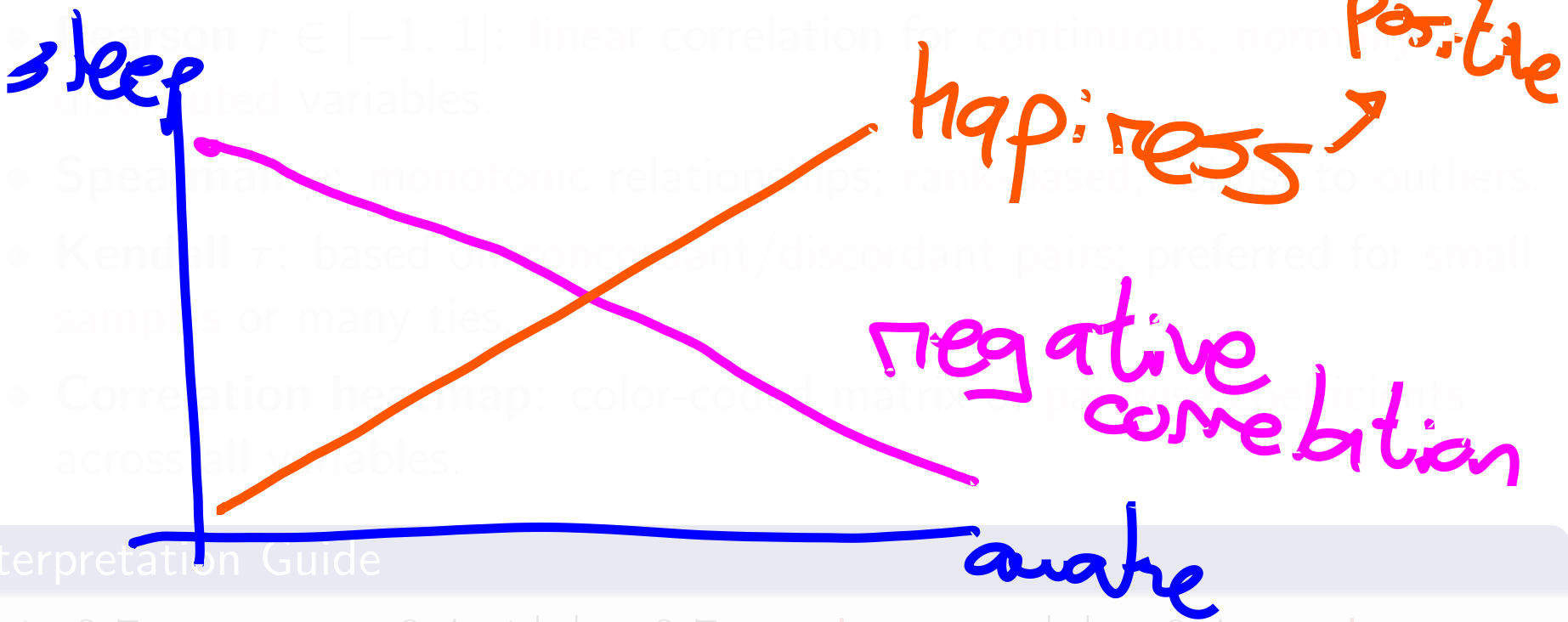
BoxPlots and Data Confidence II



Correlation Analysis

Correlation

Quantifies the **strength** and **direction** of the statistical relationship between two variables. It describes *association*, not causation



Interpretation Guide

$|r| \geq 0.7$: strong. $0.4 \leq |r| < 0.7$: moderate. $|r| < 0.4$: weak or negligible.

Correlation Analysis

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matriz

blue - red

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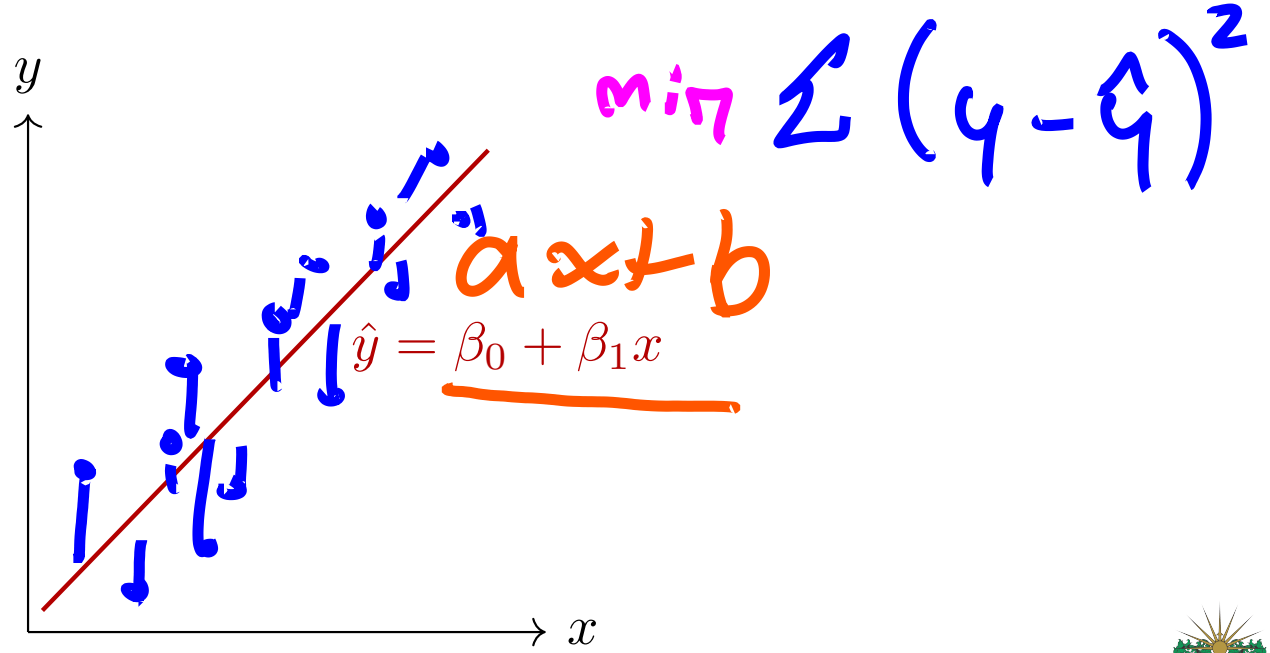
interesting

Tendency Analysis: Linear Regression I

Simple Linear Regression

Linear regression models the **linear relationship** between a dependent variable y and an independent variable x:

$$\hat{y} = \beta_0 + \beta_1 x$$







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Tendency Analysis: Linear Regression II

OLS Coefficient Estimates:

$$\beta_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \quad \beta_0 = \bar{y} - \beta_1 \bar{x}$$

Goodness of Fit:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}, \quad SS_{\text{res}} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$





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2 Data Analysis





Rust for Data Analysis I

Rust Language

Rust is a **systems programming language** focused on memory safety, zero-cost abstractions, and performance, achieving `C-level speed` without a garbage collector via its *ownership model*.

Why Rust for data?

- **No runtime overhead:** ideal for large-scale data crunching.
- **Fearless concurrency:** safe parallel processing with no data races guaranteed at compile time.
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Rust for Data Analysis II

Key crates (libraries):

- `polars`: fast DataFrame library backed by Arrow.
- `arrow2`: Apache Arrow in-memory columnar format.
- `ndarray`: N-dimensional arrays similar to NumPy.
- `rayon`: data-parallel iterators with work-stealing.





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Missing Values

Missing Data

Occur when **no data** is recorded for a variable in an observation. They can bias results, reduce statistical power, and cause **errors** in models if not handled properly.

- MCAR — *Missing Completely at Random*: absence is independent of any variable; safest to impute or drop.
- MAR — *Missing at Random*: missingness depends on other observed variables (e.g., age missing more often for elderly).
- MNAR — *Missing Not at Random*: missingness depends on the unobserved value itself (e.g., high earners skip income).





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Outlier Detection: IQR and Z-Score

IQR Method

$$IQR = Q_3 - Q_1$$

Bounds:

$$L = Q_1 - 1.5 \cdot IQR$$

$$U = Q_3 + 1.5 \cdot IQR$$

Observations outside $[L,\ U]$ are outliers.
Robust: does **not** assume normality.

Z-Score Method

$$z_i = \frac{x_i - \mu}{\sigma}$$

Values with $|z_i| > 3$ exceed 3 standard deviations and are flagged as outliers.
Assumes: data is **approximately normal**.

When to use which?

Use IQR for **skewed** or **non-normal** data. Use Z-score when data is **approximately normal** and the mean and standard deviation are reliable.

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Data Normalization

Feature Scaling

Normalization transforms features to a **common scale**, preventing `magnitude-dominated` behavior in algorithms sensitive to feature scale such as *KNN*, *SVM*, and *PCA*.

Min-Max Scaling

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Output range: $[0,\ 1]$.
Sensitive to outliers.

Z-Score Standardization

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Output: $\mu' = 0,\ \sigma' = 1$.
Robust for **normal** distributions.





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Pearson Correlation Coefficient I

Pearson's r

Pearson's r measures the **linear** association between two continuous variables X and Y, ranging from -1 to $+1$:

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Pearson Correlation Coefficient II

- **Significance test**: $t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$ follows a t-distribution with $n-2$ *degrees of freedom*.
- **Assumptions**: *linearity*, *bivariate normality*, *no outliers*, and *homoscedasticity*.





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Polars: Modern DataFrame Library I

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Advantages over pandas:

- Lazy evaluation: builds a query plan before execution, enabling automatic optimization.
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Polars: Modern DataFrame Library II

Key operations:

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- `.filter()` / `.select()`: efficient predicate pushdown to the storage layer.
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Thanks!

Questions?



Repo: *https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming*





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